
Anti-Overcharge Power Brick

Colin Frazer, Jack Tasick, Cameron Wilding

Problem Statement

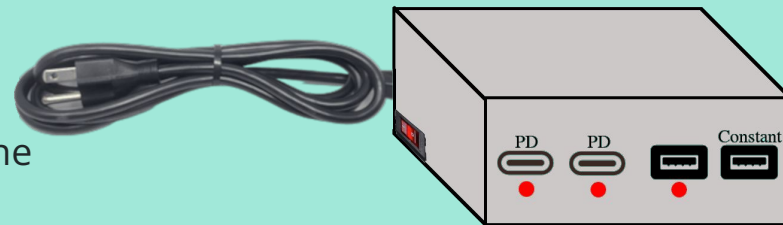
Overcharging batteries causes loss of battery capacity, overheating, and potentially leaks.

Solution: A Smart Charging Block that stops charging once a connected battery is full, extending the life of charging devices

Market: Anyone who charges devices regularly

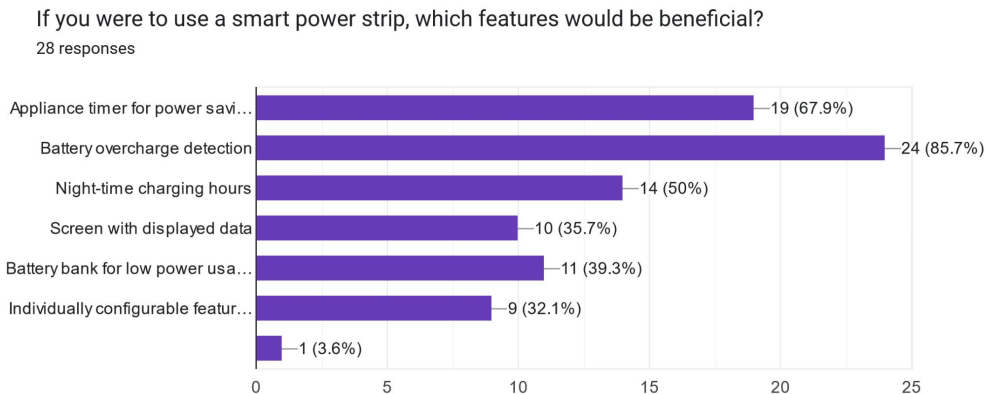
Smart Charging Block Overview

- Battery overcharge detection and auto power off
- Specifications
 - Accepts international AC input power
 - Outputs up to 60W USB-PD, 168W MAX
 - Two USB-C output ports
 - Two USB-A output ports, one controlled, one always on
 - On/Off Switch
 - Status LEDs
 - Current sensors toggle when overcharging



Customer Requirements

- Battery overcharge detection over everything else
- Portability including small size, lack of heat, etc.
- Standard outlets to support most devices, Including USB C and USB A
- Safe and reliable



Competitors



ⓘ Anker 100W USB C Charger Block (GaN II), 3 Port Fast Compact Wall Charger for MacBook, Dell XPS, iPad Pro, Galaxy S24, iPhone 16/15 and More Series

Visit the Anker Store

4.7 ★★★★★ (4,840) | Search this page

Amazon's Choice

1 sustainability feature ▾

2K+ bought in past month

\$49⁹⁹

✓prime One-Day
FREE Returns ▾

Unlock a \$100 Amazon Gift Card upon approval for Prime Visa.

Size: 1 pack

1 pack 1 pack & 1 cable



ⓘ Apple 96W USB-C Power Adapter

Visit the Apple Store

4.7 ★★★★★ (125) | Search this page

Amazon's Choice

1K+ bought in past month

-11% \$69⁹⁸

Typical price: \$79.00

Or \$12.92/mo (6 mo). Select from 1 plan

✓prime Overnight
FREE Returns ▾

Apply now and get a \$20 Amazon Gift Card upon approval of the Prime Store Card, or see if you pre-qualify with no impact to your credit bureau score.

May be available at a lower price from other sellers, potentially without free Prime shipping.

Brand Apple

Connectivity USB

Technology USB Type C

Connector Type USB Type C

Compatible Devices MacBook Air (15-inch, M2, 2023), MacBook Air (13-inch, M2, 2022), MacBook Air (13-inch, M1, 2020), MacBook Air (Retina, 13-inch, 2020), MacBook Air (Retina, 13-inc...

See more ▾

▾ See more

About this item

- The 96W USB-C Power Adapter offers fast, efficient charging at home, in the office or on the go.



Charger for Asus Laptop, (UL Certified Safety), Vivobook Zenbook and More, Featuring Charging Indicator, Foldable Plug(Scratch-Resistant), Detachable Connector, 7.5Ft Cable, 65W/45W/33W

Visit the Nicpower Store

4.4 ★★★★★ (1,818) | Search this page

500+ bought in past month

-18% \$20⁹⁹

List Price: \$25.09

✓prime One-Day
FREE Returns ▾

Unlock a \$100 Amazon Gift Card upon approval for Prime Visa.

Brand Nicpower

Connectivity powerline

Technology

Connector Type Round Connector

Compatible Devices Asus Laptop Charger 65W

Included 1 x Replacement Charger for Asus Laptop, 4 x Detachable Round Connector

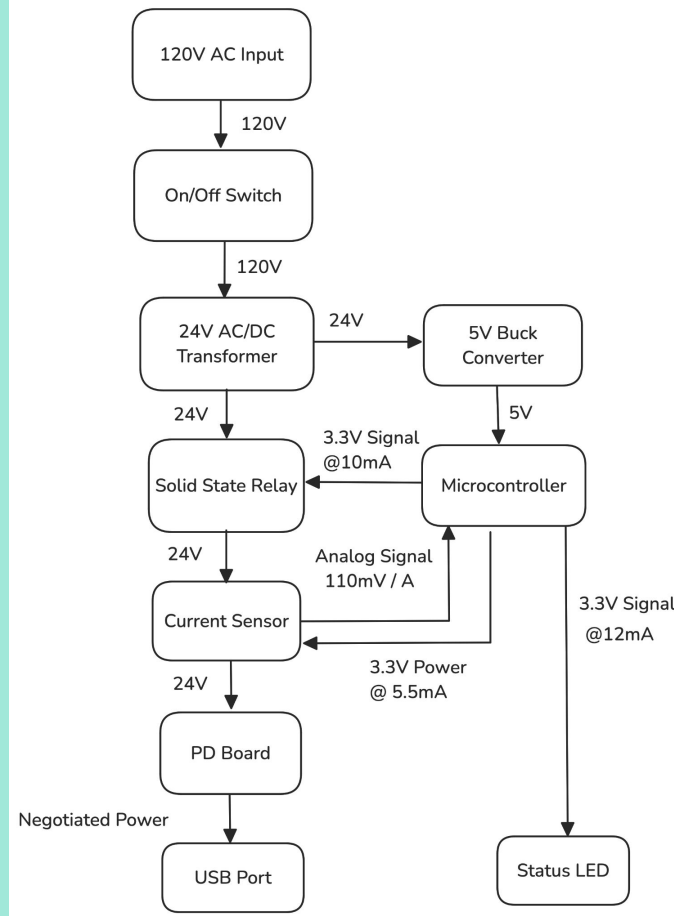
✓ See more

Value Analysis

Criterion	Anker	Apple	Asus	Ours	Weight
Ports	7	4	1	10	1
Speed	10	8	8	10	2
Overcharge	5	1	1	10	4
Price	5	3	10	1	4
Unweighted Sum	27	16	20	31	
Weighted Sum	67	36	61	74	

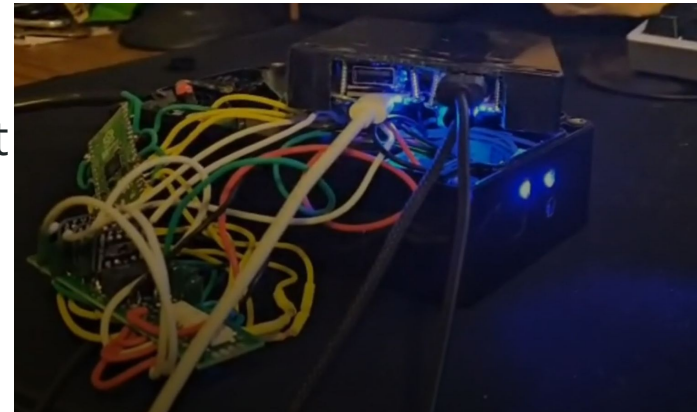
- Anker is a strong competitor
- Our overcharge feature is unique
- Purpose-bought item

Block Diagram



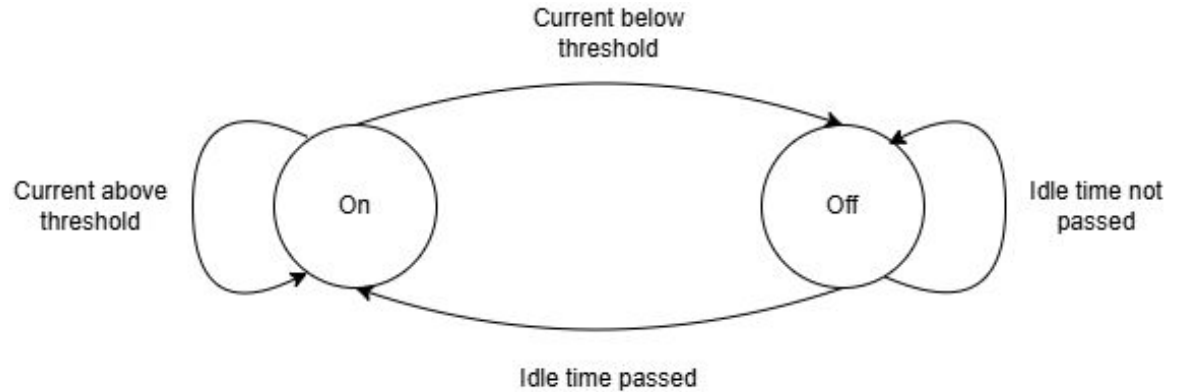
Enclosure Design and Construction

- ABS Plastic electrical box
- Final product would be 40-60% smaller
- Every connection is soldered
- Ports and Transformer are screwed down
- Rigid, has been dropped several times on cement
- Still works!

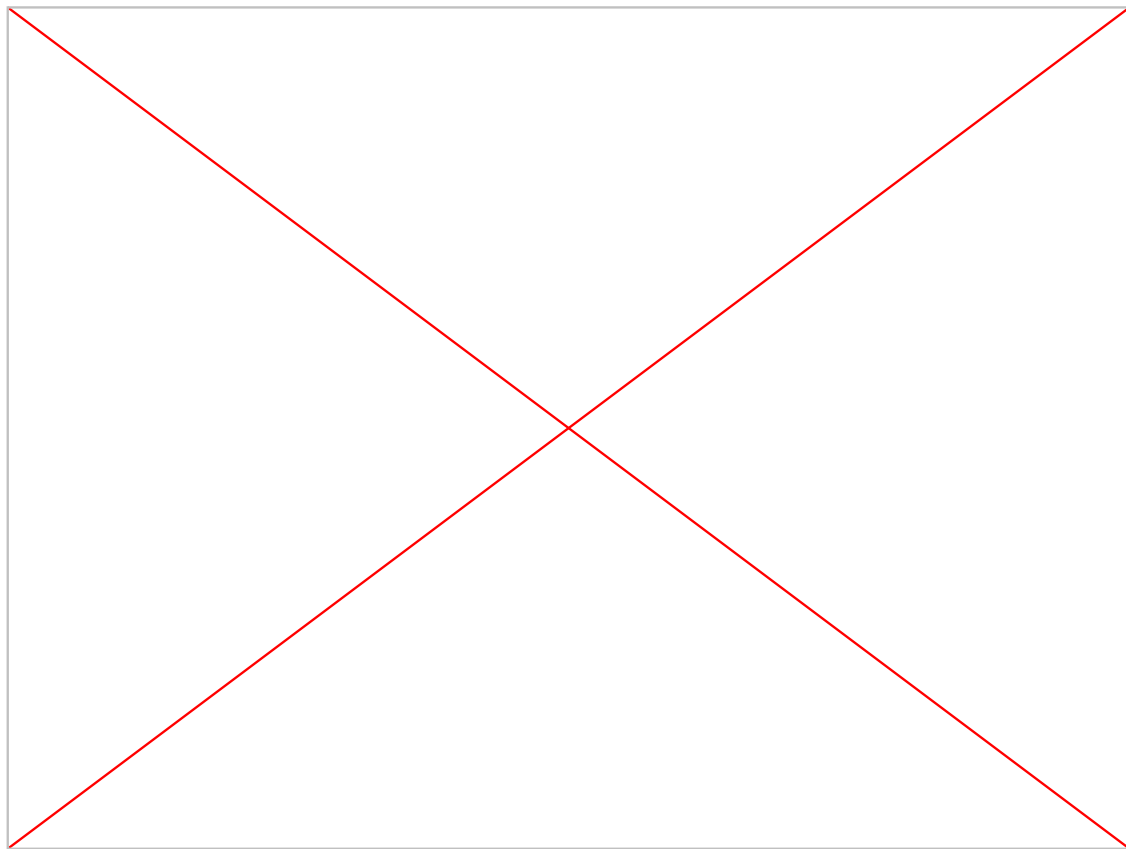


Software Functionality

- Functionality
 - Timing
 - Sensing
 - Power shut-off
 - Lighting
 - Multiple ports
- Operation
 - Hits vs idles
 - Cutoff value
 - Filtering and averaging



Demo Video



Fixed Cost Determination

- Personnel
 - 2 Engineers (1)
 - 1 Technician (2)
 - 1 Admin (3)
- Other Costs
 - Equipment/Utilities
 - Rent (4)
 - Marketing (5)
- Startup Costs
 - Marketing Materials
 - Initial Equipment

Costs		Per Month	Startup
2 Entry Level Engineers		\$15,722	
1 Technician		\$5,800	
1 Admin		\$4,500	
Rent \$18.50/ sq.ft. * 2000/yr		\$3,083.33	
Equipment/Utilities		\$1,500	
Marketing		\$1,000	
Startup Equipment			\$8,000
Marketing Materials			\$1,000
Total Per Month		\$31,605	
Total Startup			\$9,000

Variable Cost Determination

- Part Costs
 - Reduced ~70% from prototype
- Assembly Cost
 - \$0.01-0.05 per component, estimated 90 components (6)
- Test Cost
- Packaging Cost
- Amazon Fees (7)

Parts	\$34.36
Assembly Cost	\$3.50
Test Cost (5 min per unit)	\$4.00
Packaging Cost	\$1.11
Bulk Shipping to Amazon	\$4.00
Sum	\$46.97
Amazon Fees (20%)	\$25.29
Total Cost per Device:	\$72.26
Selling Price (175%)	\$126.46

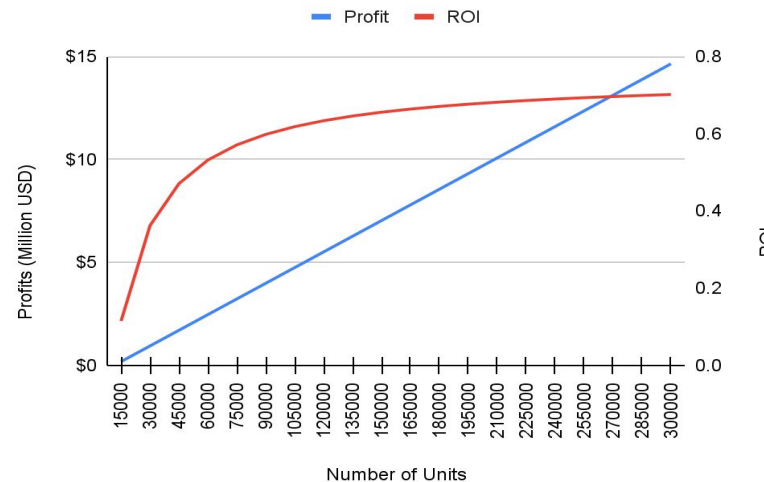
Return on Investment

ROI Breakeven Analysis for June, Year 2



Breakeven Point: 10663 Units

ROI VS Number of Units



ROI: 0.75

Prototype Functionality

- Current sensors lack precision, have consistent accuracy
- Difficult to detect small changes
- Analog filtering and scaling would fix this
- Tests showed that devices from headphones to laptops toggle effectively

Battery capacity range tested:

Earbuds: ~2Wh

Laptop: 71Wh



1.99 Wh (8)



71 Wh (9)

Sources

- (1) <https://www.ziprecruiter.com/Salaries/What-Is-the-Average-Entry-Level-Electrical-Engineer-Salary-by-State>
- (2) <https://www.indeed.com/career/electrical-technician/salaries>
- (3) <https://www.salary.com/research/salary/recruiting/administrator-salary>
- (4) <https://www.commercialcafe.com/commercial-property/us/ma/worcester/200-lincoln-street/>
- (5) <https://www.webfx.com/digital-marketing/pricing/cost-of-marketing/>
- (6) <https://www.chinapcbassembly.com/blog/detailed-pcb-assembly-cost-breakdown/>
- (7) https://sellercentral.amazon.com/revcalpublic?ref_=sdus_pri_xscus_rc2&IdStackingCodes=ELUSSOA-issues.amazon.com%3EELUSSOA-app.asana.com%3EELXXSOA-app.asana.com%3ESCUSWPDirect%3EELXXSOA-app.asana.com&initialSessionID=134-7516809-1571051&Id=ELUSSOA-duckgo.com&pageName=US%3ASD%3ASOA-pricing-estimate
- (8) <https://www.amazon.com/IFRESEM-Replacement-Battery-Charging-EP-QR510/dp/B0CNZ6GGJ3>
- (9) <https://www.amazon.com/AURUNHO-L19M4PDC-Replacement-7-15ITL5-7-14ACN6-82N7/dp/B0D4QFZ84M>